

A Short Practical Introduction to Bayesian Analysis

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24–25 September 2026

A short worked example

The Bayesian toolkit

The Bayesian workflow

Working through the workflow: golf putting

Becoming independent Bayesian analysts

Course notebooks, slides, and data

github.com/opherdonchin/BayesShortCourse

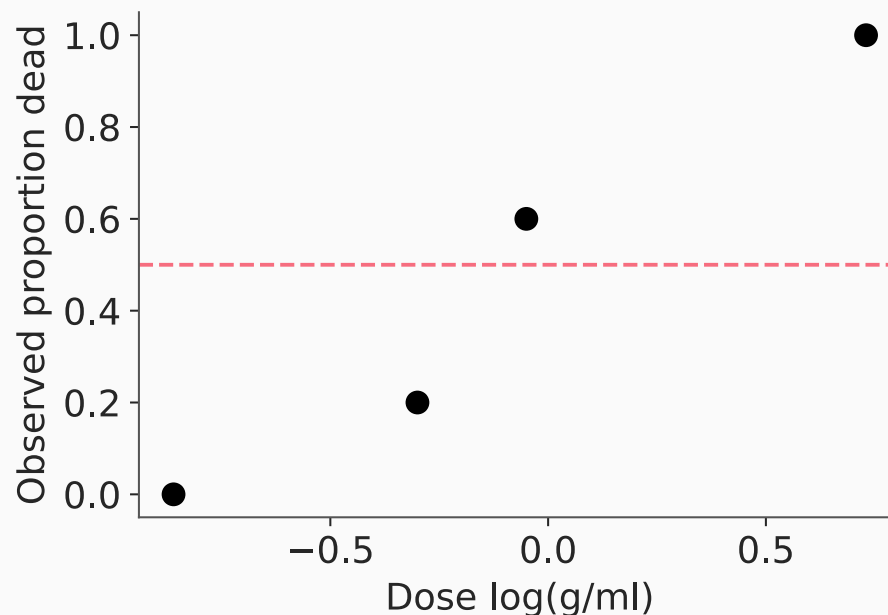
A short worked example

Dose response

Four doses, twenty animals

A toxin is given to rats at four doses, five rats per dose.

log dose	rats	deaths
-0.86	5	0
-0.30	5	1
-0.05	5	3
0.73	5	5



Data: Racine-Poon et al. (1986); Gelman & Vehtari, *Bayesian Workflow*, §3.5

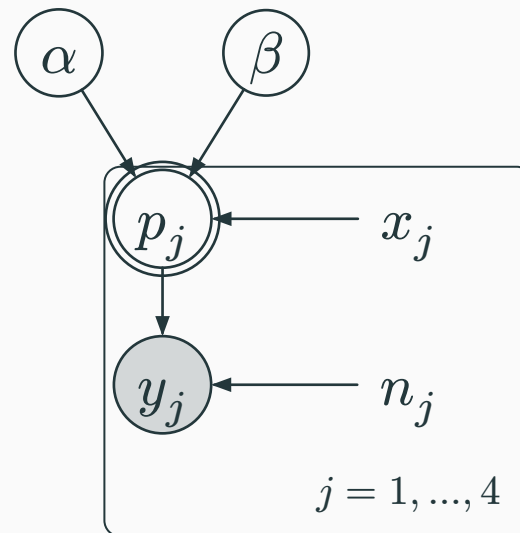
A generative model for mortality

$$y_j \sim \text{Binomial}(n_j, p_j)$$

$$\text{logit}(p_j) = \alpha + \beta x_j$$

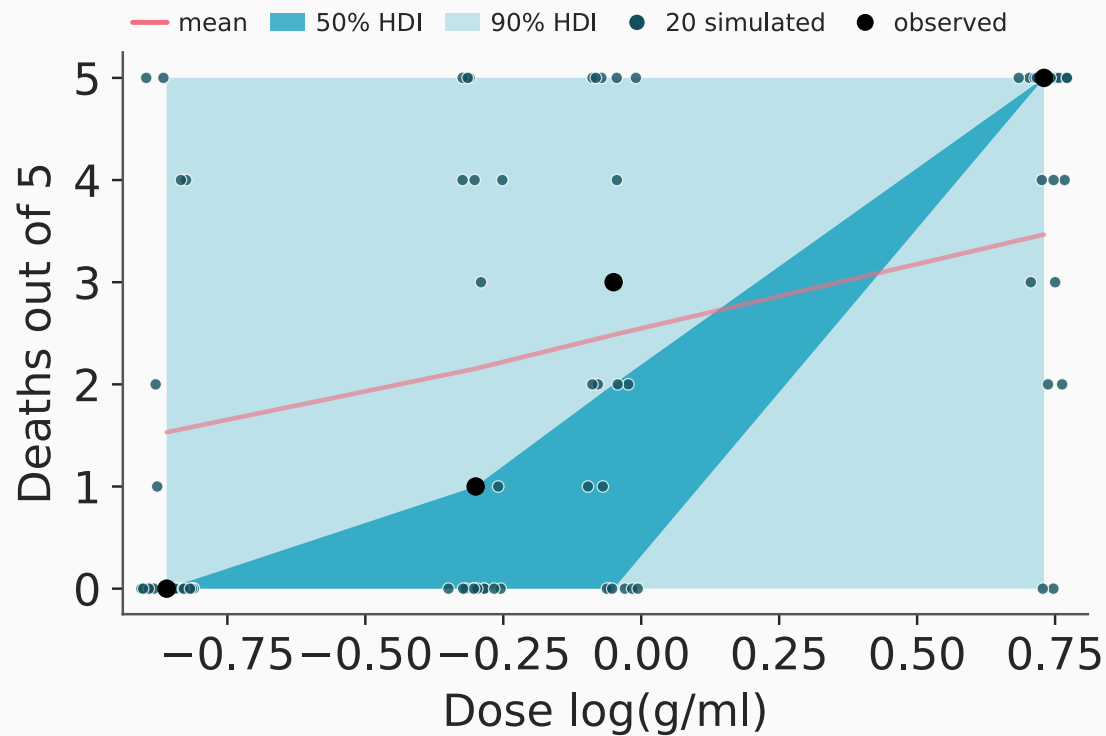
$$\alpha \sim \text{Normal}(0, 5)$$

$$\beta \sim \text{HalfNormal}(5)$$



```
with pm.Model() as model:  
    alpha = pm.Normal("alpha", mu=0, sigma=5)  
    beta = pm.HalfNormal("beta", sigma=5)  
    logit_p = alpha + beta * dose  
    pm.Binomial("deaths", n=n, logit_p=logit_p, observed=deaths)
```

Priors: what could we have seen?



Prior predictive data should not

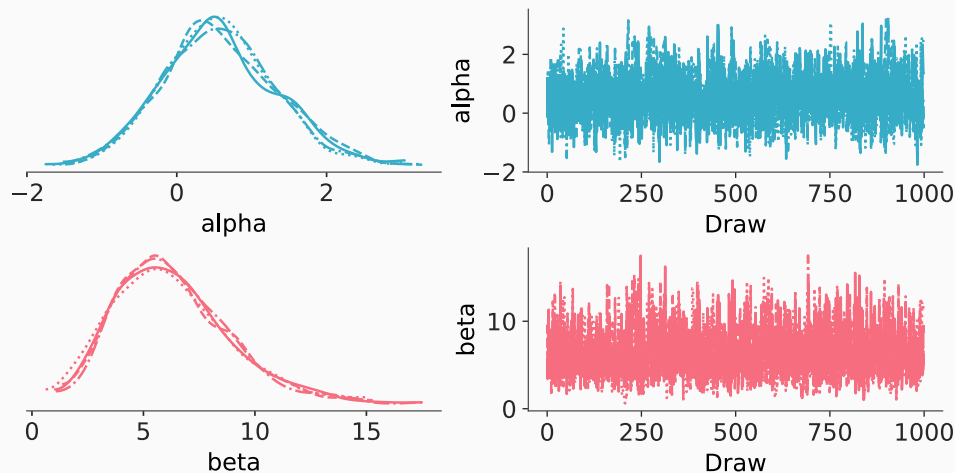
- violate common sense
- make our data highly unlikely
- confine the model to our data

Sampling and diagnostics

Fitting the model means drawing samples from the posterior distribution.

```
with model:
```

```
idata = pm.sample(draws=1000, chains=4)
```



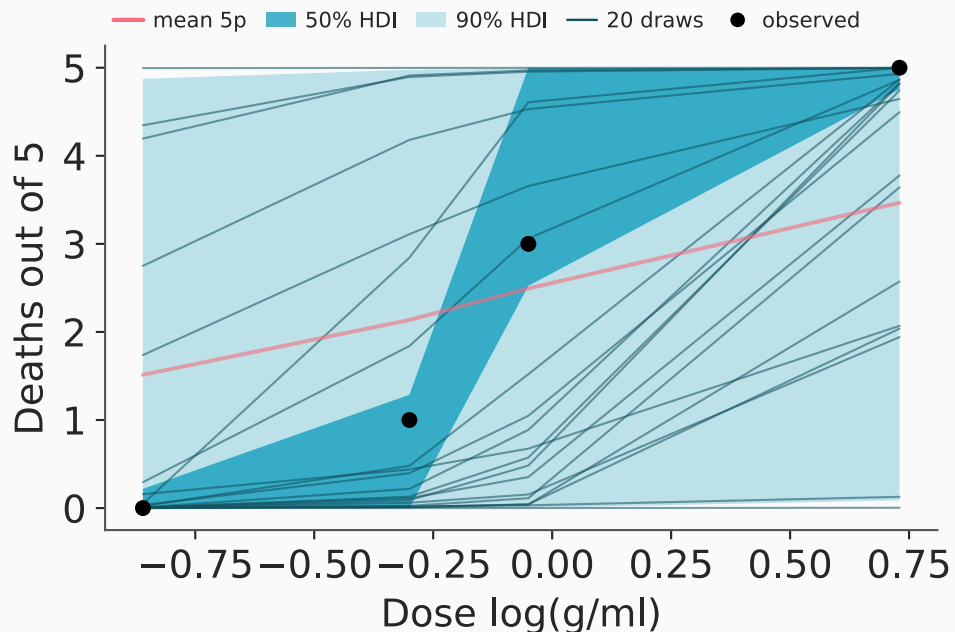
Check before interpreting:

- 0 divergences
- $\hat{R} = 1.00$
- ESS > 1,300 of 4,000 draws

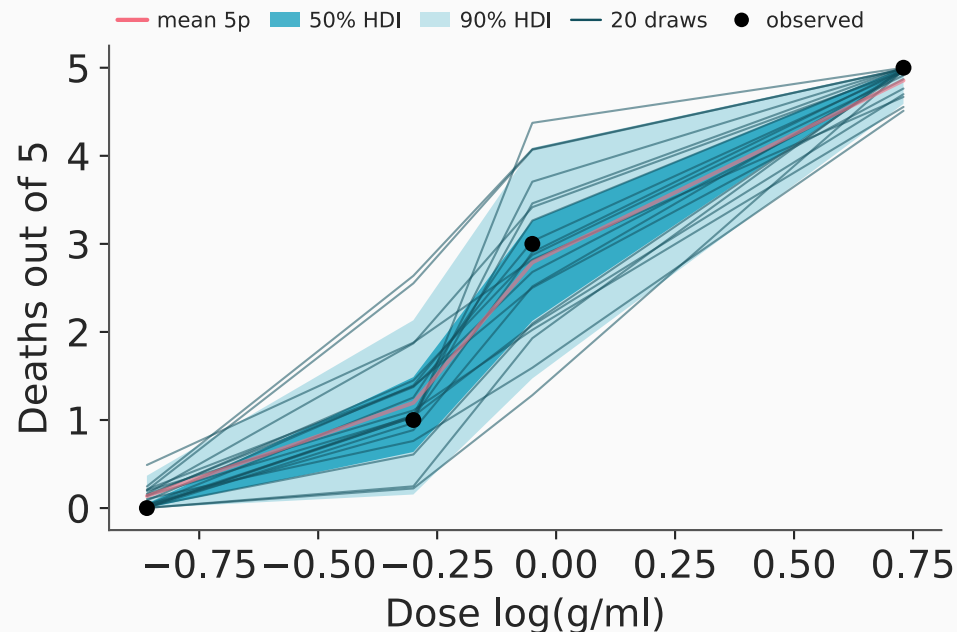
Posterior: which dose-response curves remain plausible?

The bands show uncertainty about the mean number of deaths.

Prior

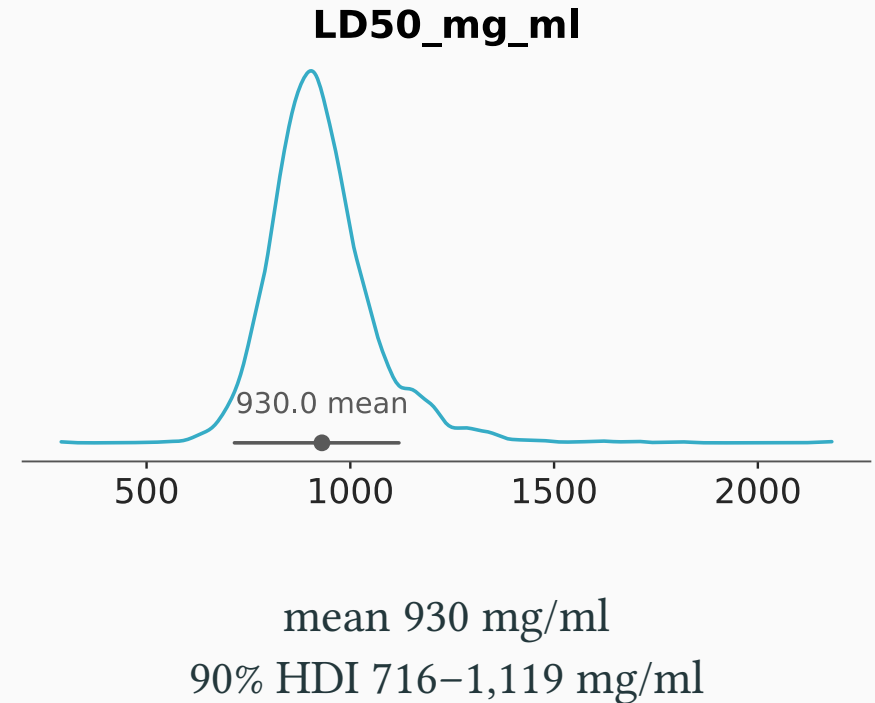


Posterior

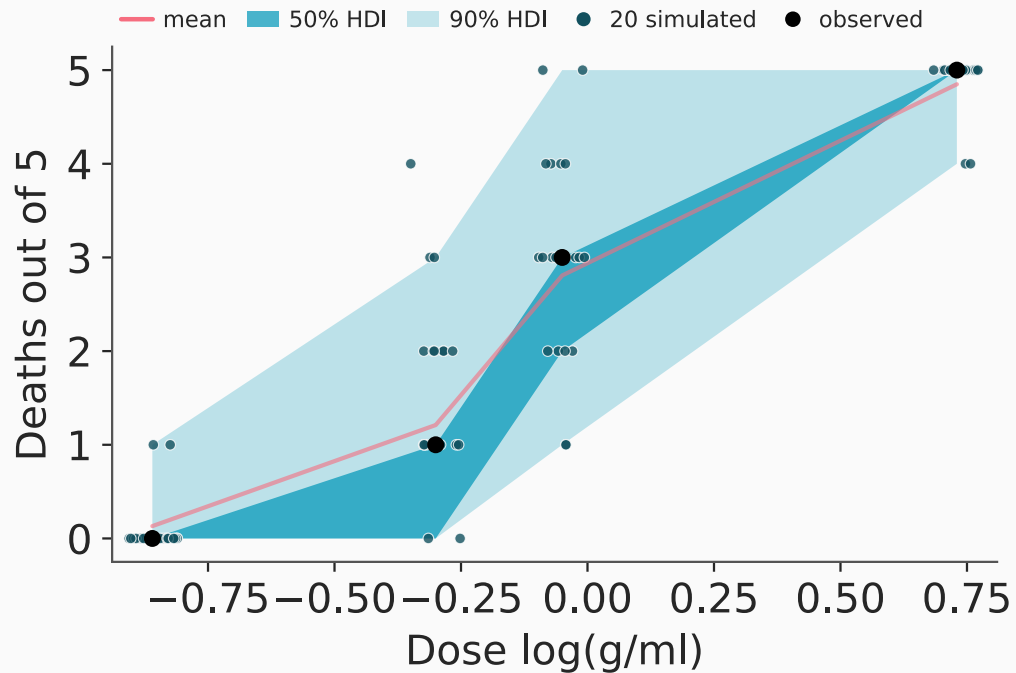


LD50: the quantity we actually care about

- Any quantity can be computed from the parameters:
 - $LD50 = -\alpha/\beta$
 - Once for every posterior draw.
- So every quantity comes with its own uncertainty.
- Here *how well* we know LD50 matters as much as its value.



Posterior predictive: can the model reproduce the experiment?



This is what the model predicts we could have seen, *after* fitting.

- A workflow check meant to be done before looking at the posterior.
- Workflow is not lockstep: going back to check is always fine.
- We understand data better than we understand parameters.

The Bayesian toolkit

Four libraries, one workflow



Write the model, sample the posterior

pymc.io



Choose and check priors

preliz.readthedocs.io



Diagnose, summarize, and plot

python.arviz.org

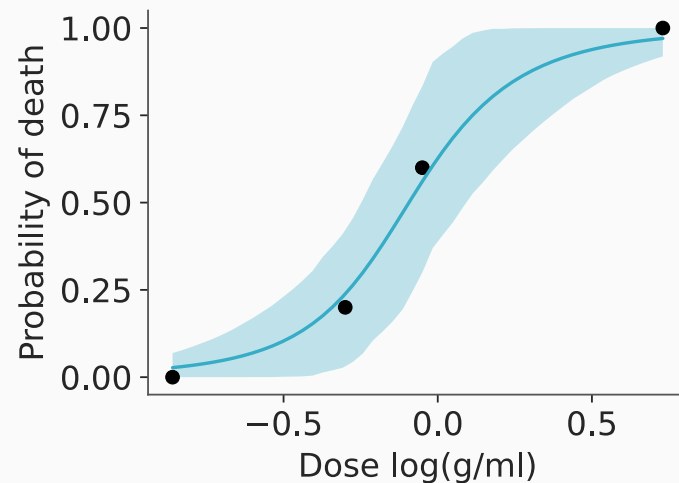


Regression models from a formula

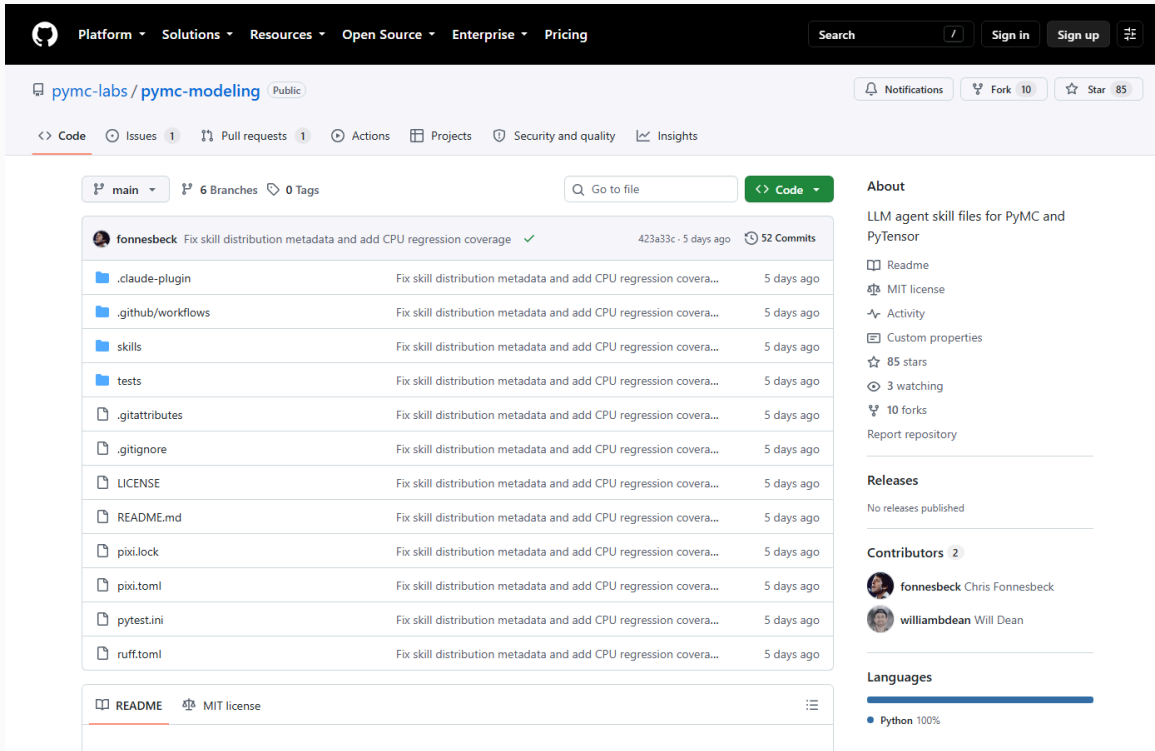
bambinos.github.io/bambi

Bambi: the same model as one formula

```
priors = {  
    "Intercept": bmb.Prior("Normal", mu=0, sigma=5),  
    "dose": bmb.Prior("HalfNormal", sigma=5),  
}  
model = bmb.Model("p(deaths, n) ~ dose", data,  
                  family="binomial", priors=priors)  
idata = model.fit()  
bmb.interpret.plot_predictions(model, idata, "dose")
```



Coding assistants can learn current PyMC practice

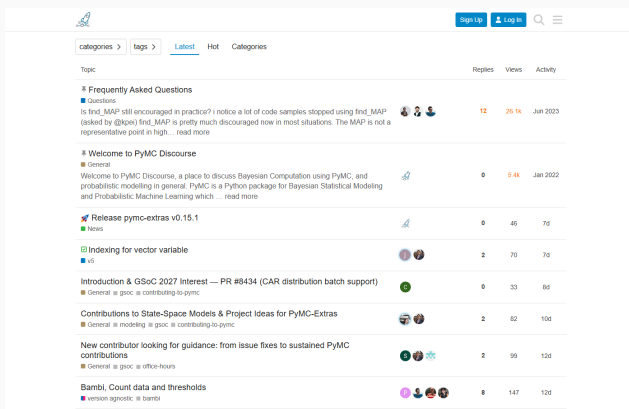


github.com/pymc-labs/pymc-modeling

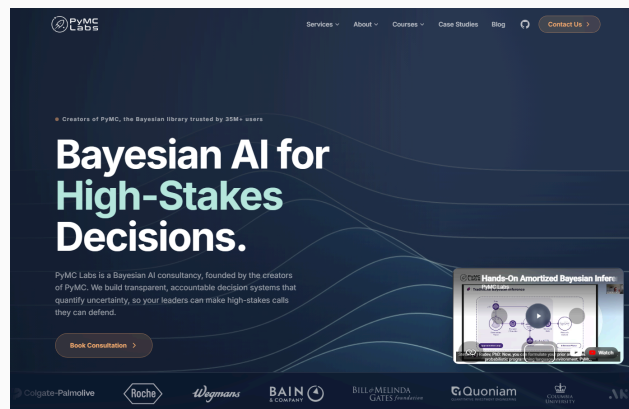
Skill files from PyMC Labs that teach AI coding assistants current PyMC and ArviZ practice.

We used them to build this course's notebooks.

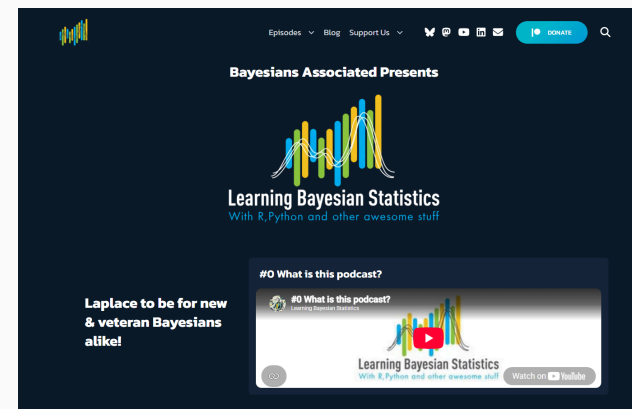
Where to ask, hire, and listen



Ask: PyMC Discourse
discourse.pymc.io



Hire: PyMC Labs
pymc-labs.com



Listen: Learning Bayesian Statistics
learnbayesstats.com

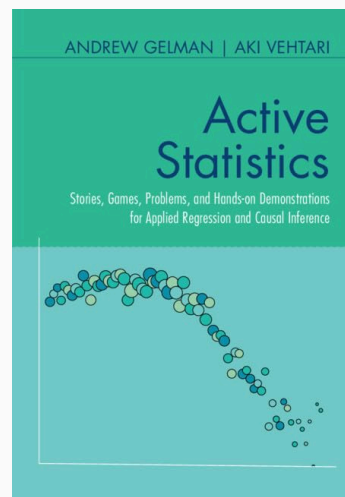
Books worth reading next



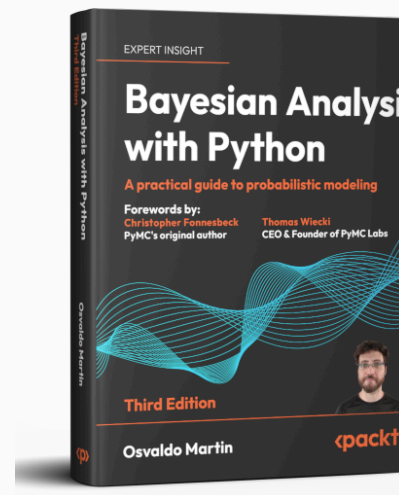
McElreath
Statistical Rethinking



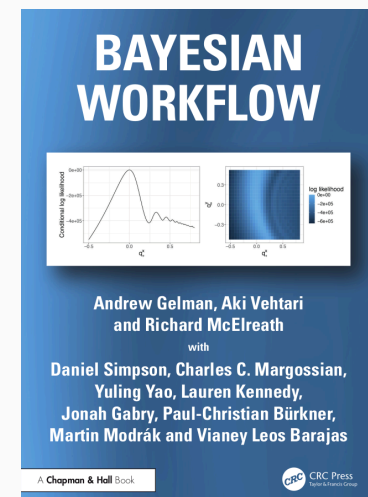
Gelman, Hill & Vehtari
Regression and Other Stories



Gelman & Vehtari
Active Statistics



Martin
Bayesian Analysis with Python



Gelman, Vehtari et al.
Bayesian Workflow

The Bayesian workflow

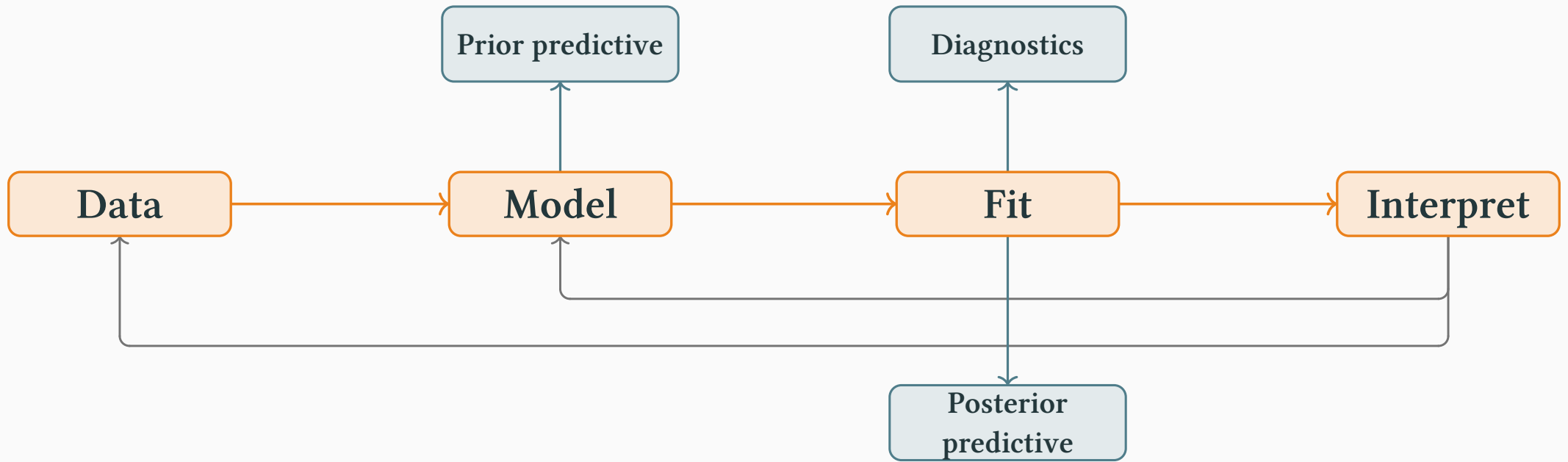
The workflow starts with a four-step loop



Even at this level, we allow for

different interpretations · iterative experimental design

Checks make each transition accountable



Validation belongs at each step

Prior predictive

Diagnostics

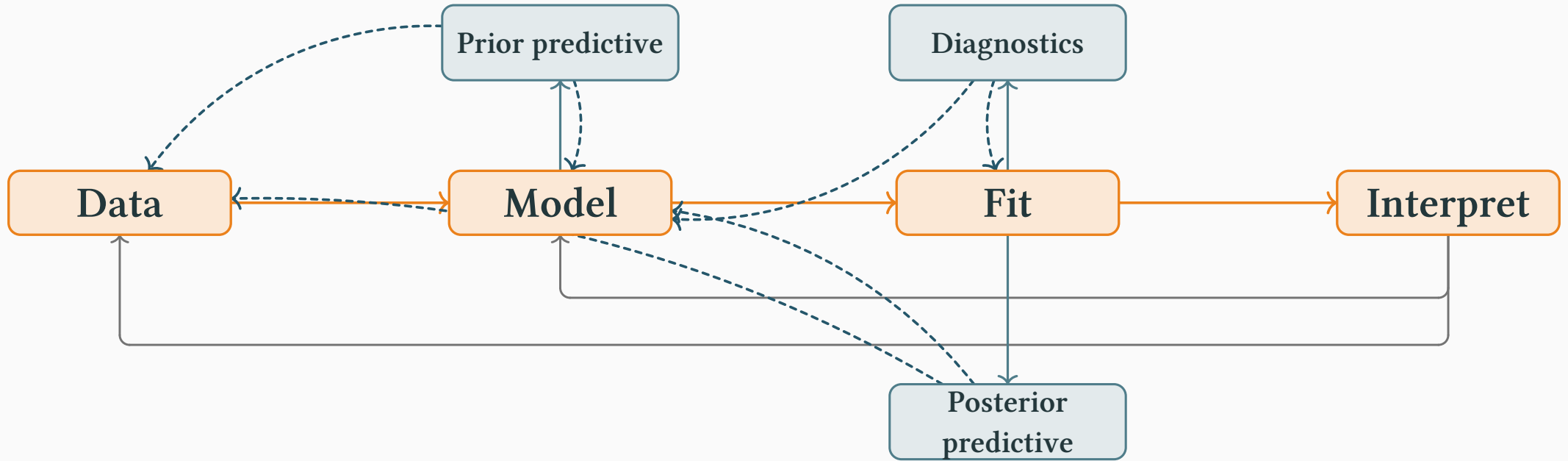
Posterior predictive

tests model coherence

test whether fitting converged

tests fit to the observed data

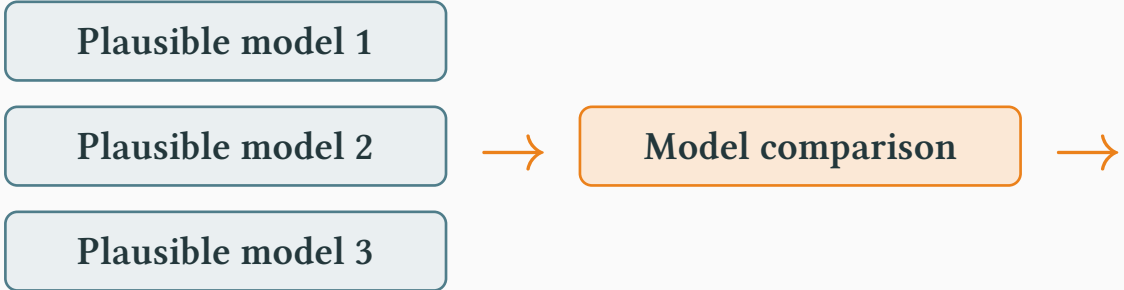
A failed check tells us where to return



A failed check
Prior predictive
Diagnostics
Posterior predictive

means something must change
revisit assumptions about the model or data
revisit the model or the computation
reassess how the model represents the data

Model comparison asks which useful story predicts best



Leave-one-out cross-validation			
Model	LOO	SE	Δ LOO
Model 1	-124.8	4.5	13.0
Model 2	-112.4	0.3	0.6
Model 3	-111.8	-	0.0

illustrative values; larger LOO is better

The workflow can lead to several **plausible models** with different interpretations.
It is also a useful way to think about frequentist “null models.”
Model comparison is a basic tool in Bayesian data analysis.
All models are wrong; some models are informative.

Working through the workflow: golf putting

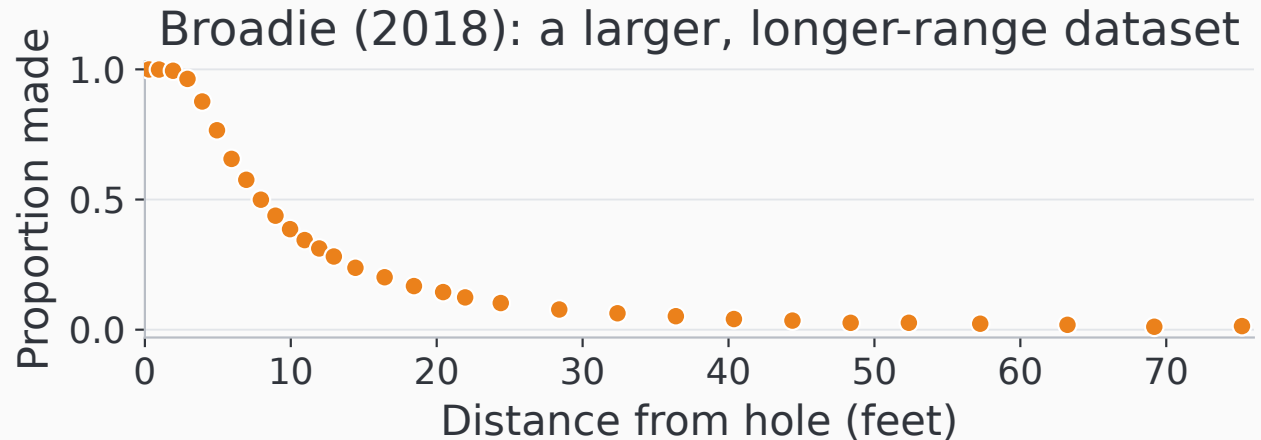
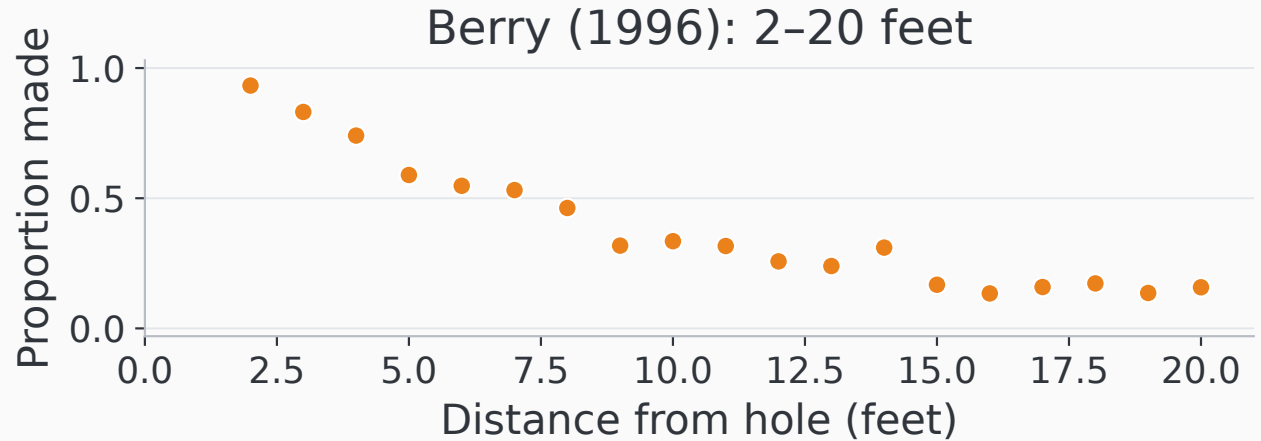
The data



AI-generated course illustration

Berry (1996): professional putting summaries at rounded distances from 2 to 20 feet.

Broadie (2018): a later, much larger dataset extending to 75 feet.

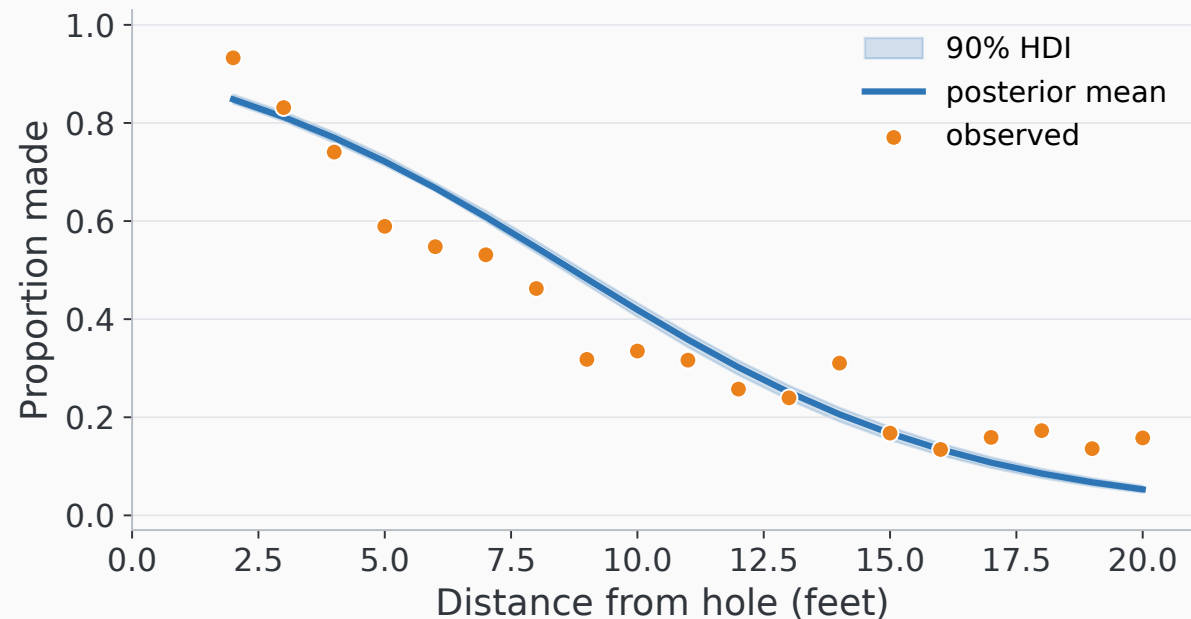


Starting with a simple model is easy and informative

$$y_j \sim \text{Binomial}(n_j, p_j)$$

$$\text{logit}(p_j) = \alpha + \beta x_j$$

```
with pm.Model() as model:
    alpha = pm.Normal("alpha", 0, 3)
    beta = pm.Normal("beta", 0, 0.5)
    logit_p = alpha + beta *
distance
    pm.Binomial(
        "made", n=attempts,
        logit_p=logit_p,
        observed=made,
    )
```



Prior and posterior predictive checks develop intuition for what it can generate.

What it does and does not capture tells us how much remains to explain.

Geometry turns distance into a success probability



Redrawn from Gelman et al., *Bayesian Workflow*, Fig. 25.3

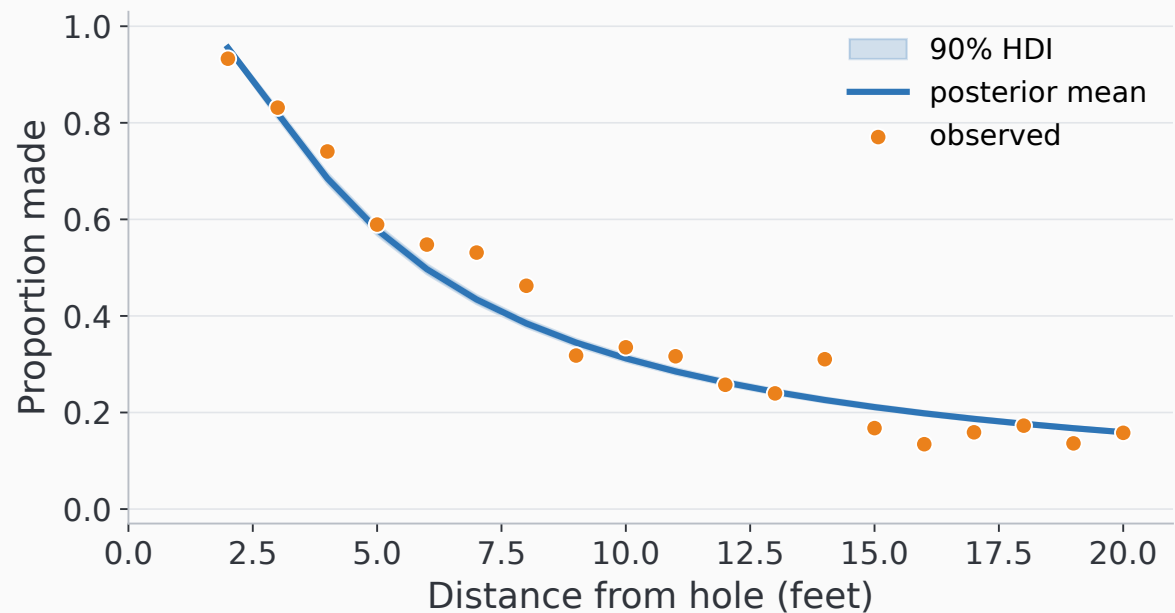
$$\theta(x) = \arcsin\left(\frac{R-r}{x}\right)$$

$$p(x) = 2\Phi\left(\frac{\theta(x)}{\sigma}\right) - 1$$

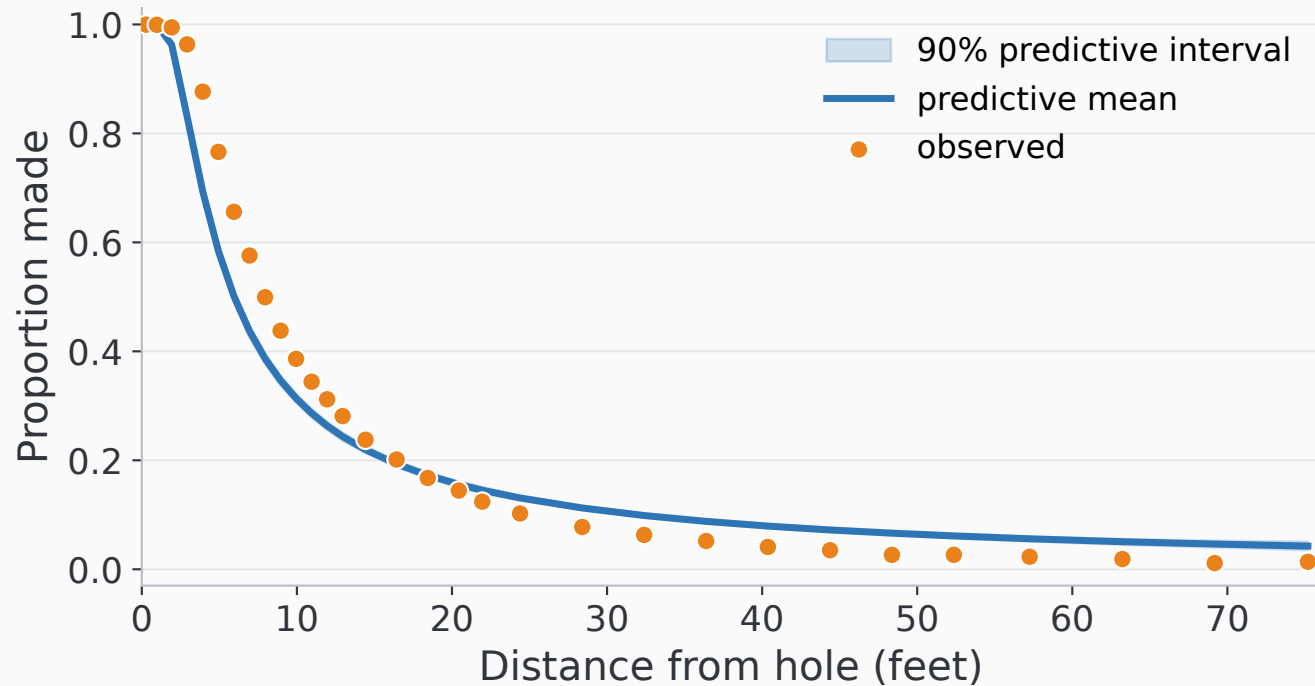
- A highly simplified model of putting physics
- The only parameter is noise in the aiming angle, σ
- These counts overwhelm almost any reasonable prior
- It is still healthy to ask what a reasonable prior implies

The aiming error model

```
with pm.Model() as model:
    sigma_deg = pm.LogNormal(
        "sigma_deg", log(2), 0.7
    )
    theta = pm.math.arcsin(
        (cup_radius - ball_radius)
        / distance
    )
    p = 2 * pm.math.invprobit(
        theta / deg2rad(sigma_deg)
    ) - 1
    pm.Binomial(
        "made", n=attempts,
        p=p, observed=made,
    )
```



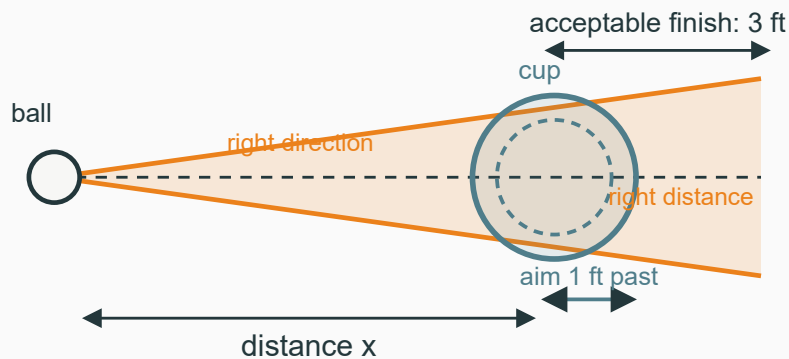
The aiming error model on new data



When we have faith in a model, we can test it against new data as those data become available.

Data: Broadie (2018); comparison follows *Bayesian Workflow*, §25.3

A new physical model



Redrawn from Gelman et al., *Bayesian Workflow*, Fig. 25.6

Direction

$$p_{\text{angle}(x)} = 2\Phi\left(\frac{\arcsin\left(\frac{R-r}{x}\right)}{\sigma_{\text{angle}}}\right) - 1$$

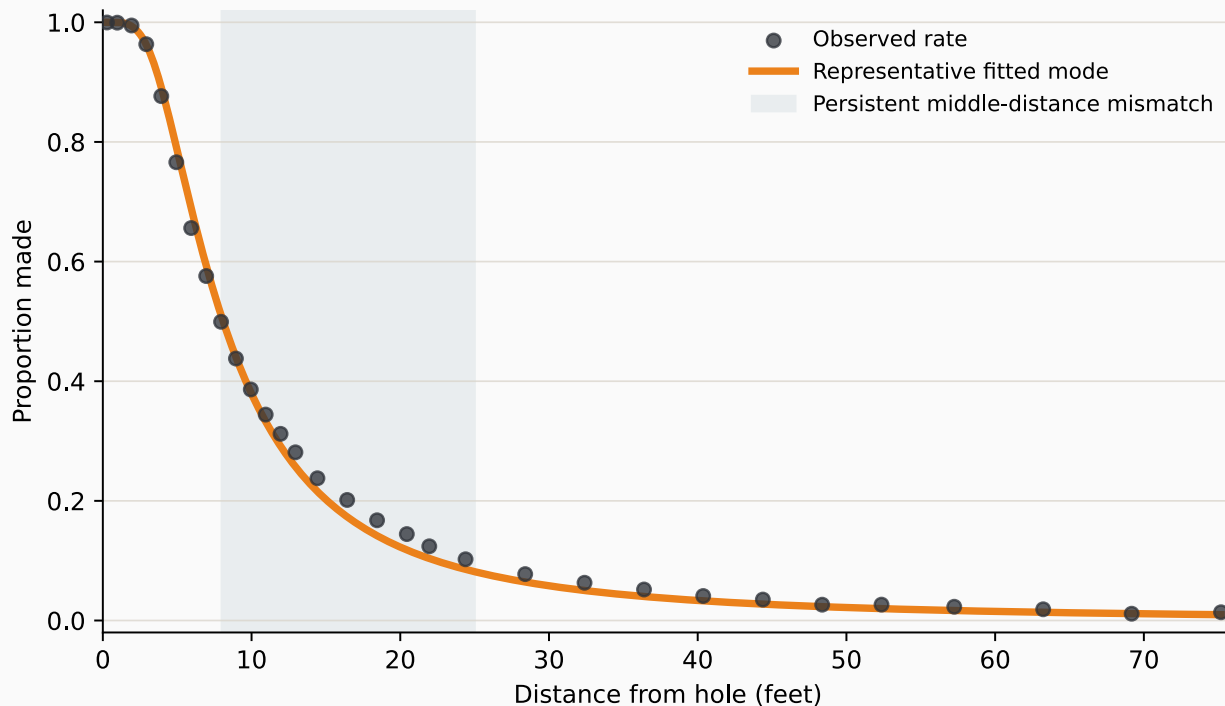
Distance

$$u = (x + 1)(1 + \varepsilon), \quad \varepsilon \sim \text{Normal}(0, \sigma_{\text{distance}})$$

$$p_{\text{distance}(x)} = \Phi\left(\frac{2}{(x + 1)\sigma_{\text{distance}}}\right) - \Phi\left(-\frac{1}{(x + 1)\sigma_{\text{distance}}}\right)$$

$$p(x) = p_{\text{angle}(x)}p_{\text{distance}(x)}$$

The aiming + distance model

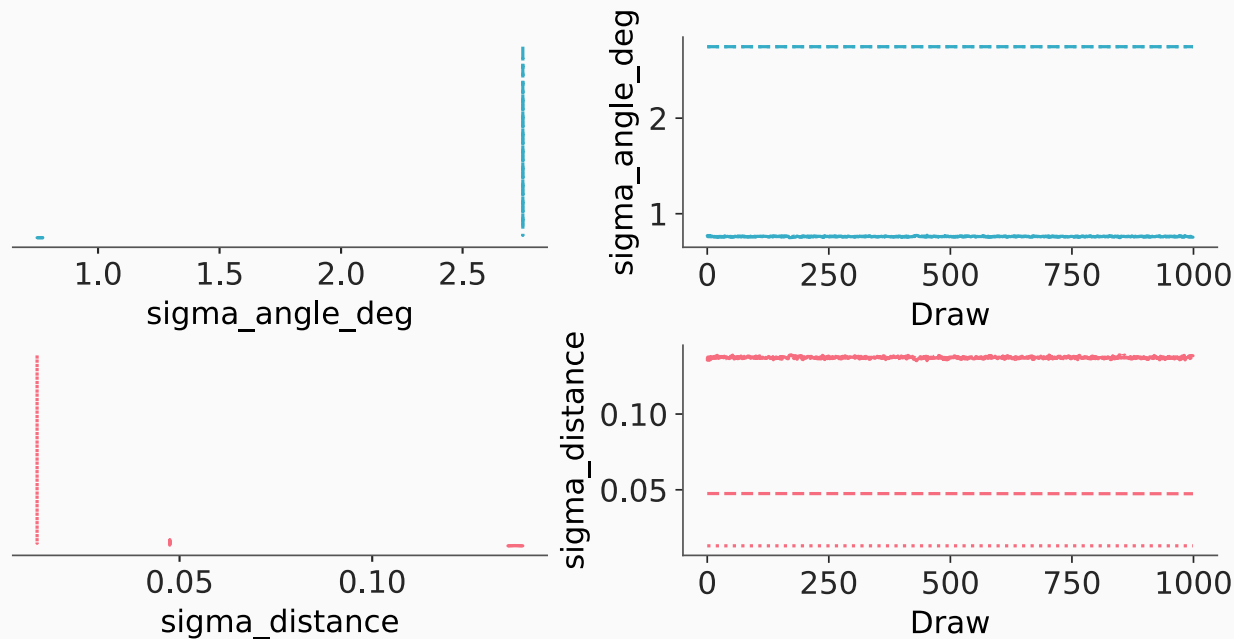


An aiming and distance model improves fit and interpretability.

Still: consistent fit problems remain at middle distances.

A plausible curve is not yet a trustworthy posterior.

The aiming + distance model: have we sampled the posterior?



The chains disagree:

- $\hat{R} \approx 2$
- ESS ≈ 5 of 4,000 draws

We have not sampled the posterior.

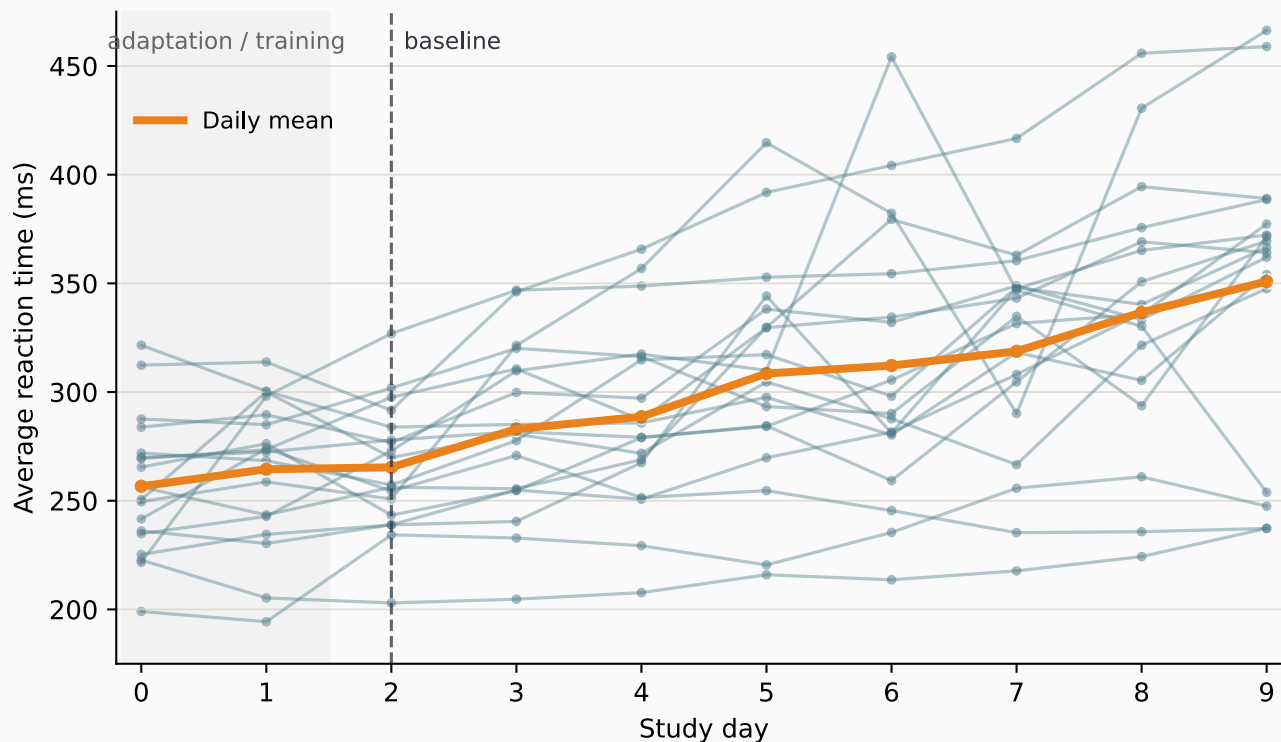
So model fit is not even the question yet.

Each failed check suggests the next model—not the final model

- **Absorb local mismatch.** Huge short-putt counts dominate on the logit scale. Replace the Binomial with a Normal discrepancy model: **wrong, but better.**
- **Try a “truer” correction.** Add distance-dependent variance on the logit scale. The fit becomes **much worse**: realism alone does not guarantee usefulness.
- **Change the discrepancy scale.** Use proportional rather than additive noise. The mechanism becomes usable enough to learn the distance-tolerance parameter.
- **Keep iterating critically.** Model building develops intuition—and the habit of asking what each apparent improvement actually explains.

Becoming independent Bayesian analysts

The sleep deprivation data



180 observations

- 18 participants
- 10 daily averages each

Most sleep-restricted group

- 3 hours time in bed per night
- Days 0–1: adaptation and training
- Day 2: baseline
- Restriction begins after day 2

The population trend is clear—yet participants differ in both baseline and change.

Data: Belenky et al. (2003), distributed as `sleepstudy` in `lme4`

What we will practice

01

Choose priors

- Use common sense
- Check prior predictions
- Handle practical constraints

02

Build hierarchical models

- Separate individual and population uncertainty
- See how partial pooling shares information
- Choose priors at both levels

03

Use AI as a collaborator

- Make assumptions explicit
- Check generated code and results
- Keep scientific judgment in the loop

The goal is independence: formulate, fit, check, revise, and explain.